Ananya Deepak Deoghare

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Experience

Senior Associate Algorithm Engineer Jul 2026 to Present
QuinStreet *Foster City, CA*

- Technical owner of the end to end ML ranking architecture for the Personal Loans vertical (multi-million-dollar revenue line); scope expanded in 2026 to include the conversion model alongside CTR models, owning ranking performance, deployment strategy, experimentation, and long-term system scalability.
- Built and productionized a conversion prediction model for the Personal Loans funnel, complementing CTR models to optimize downstream conversion.

Associate Algorithm Engineer Oct 2023 to Jul 2026
QuinStreet *Foster City, CA*

- Designed and optimized machine learning models for predictive modeling and ranking on a large-scale personal loans user dataset; performed in-depth exploratory data analysis to uncover behavioral patterns and drive feature strategy.
- Built and productionized Gradient Boosting Machine (GBM) models with rigorous hyperparameter tuning, cross-validation, and feature selection, improving ranking quality and predictive performance in the banking domain.
- Applied clustering and controlled A/B experimentation to optimize advertiser click-through rates, driving a projected 10% increase in user engagement.
- Led weekly model diagnostics and performance analysis on user interaction data, increasing model accuracy by 5% and reducing error rates by 10%.
- Engineered high-signal features through systematic feature extraction and selection, significantly improving model efficiency, interpretability, and generalization.
- Partnered with data engineering and product teams to deploy SEO-driven ML models to production, enabling real-time decision-making in high-impact business workflows.
- Implemented continuous model evaluation pipelines with automated hyperparameter optimization and validation frameworks to ensure robustness and alignment with business KPIs.
- Own the performance and reliability of the team's Cython simulation framework used to stress-test ranking strategies before release; optimized processing time by 50%, meeting production-grade performance requirements.
- Implemented the Hungarian Algorithm for optimal ranking assignment, benchmarking against a greedy baseline and increasing margin revenue by 5–8%; improved handling of edge cases missed by heuristic approaches.
- Performed feature analysis for the ranking models

Machine Learning Intern Jun 2022 – Sep 2022
VidMob *New York, NY*

- Built a deep learning model to score creative ad performance, improving scoring accuracy by 25%.
- Integrated predictive modeling and recommendation engines into an AI-driven ad-tech platform.
- Conducted model evaluation pipelines to assess ad engagement prediction, resulting in better client satisfaction by 5%.

Student Researcher Jul 2020 – January 2023
University of California, Los Angeles(UCLA) *Los Angeles, CA*

- Co-authored paper (SIGGRAPH 2022) on multimodal sensing using radar and RGB data for equitable rPPG.
- Proposed a Shift-Robust Loss Function for pulse prediction, decreasing error by 40%.
- Built ML pipelines to mitigate skin-tone bias using sensor fusion, improving fairness and accuracy by 75%.

Software Engineer Jun 2019 – Jul 2021
Accenture *Bengaluru, India*

- Improved pharma sales data analysis via automated data processing pipelines, reducing manual time by 50%.
- Boosted competitor match rate by 20% via advanced analytics, earning cross-region recognition.
- Participated in Accenture's Global Innovation Challenge as a semi-finalist.

Research Experience

Research Intern, Artificial Intelligence and Robotics Lab Jan 2019 to Jun 2019
Indian Institute of Science, India

- Collaborated with Dr. Sundaram and a team of researchers to pioneer new methods for faster and more accurate diagnosis of Autism, resulting in improved treatment outcomes.
- Developed and optimized deep neural networks to analyze fMRI scans for early detection of Autism, achieving an 80% accuracy rate on a large set of cleaned data.

Automatic Garbage Segregator (with Dr. Koshy George)

- Engineered a crane that could segregate waste into biodegradable, non biodegradable, and electronic waste with an accuracy of 95.18%.
- Tested various Feature Extraction techniques like PCA, LDR, and Convolutional Neural Networks.

Volunteer, IEEE Symposium Series on Computational Intelligence (IEEE SSCI 2018) Nov 2018
Bengaluru, India

- Volunteered at the 2018 IEEE Symposium Series on Computational Intelligence, held November 18 to 21, 2018, at the Sheraton Grand Bangalore Hotel at Brigade Gateway in Bengaluru, India.
- The event featured co located symposia spanning computational intelligence areas such as adaptive dynamic programming, reinforcement learning, cognitive algorithms, neural networks, evolutionary computation, and fuzzy systems, fostering cross disciplinary exchange among global researchers and students.

Technical Skills

- **Tools & Languages:** Python, SQL, Git, Cython, AWS, GCP, MATLAB, Java, C, C++, HTML, Visual Studio, JIRA, Perforce
- **Databases:** MySQL, Amazon Redshift, Apache Hive, Feature Stores, Data Pipelines
- **Libraries:** Scikit-learn, PyTorch, TensorFlow, NumPy, Pandas, SHAP, OpenCV, Scipy, Cython
- **ML Expertise:** Ranking & Recommendations, CTR/Conversion Modeling, A/B Testing, Feature Engineering, Multi Armed Bandits, Model Monitoring, MLOps

Education

University of California, Los Angeles(UCLA) Sept 2021 - Oct 2023
MS in Electrical and Computer Engineering [GPA: 3.97/4.0] *Los Angeles, CA*

PES University Aug 2015 - Aug 2019
BTech in Electronics and Communication Engineering *Bangalore, India*

Projects

- **Multi-Class EEG Motor Imagery Classification Using Deep Learning Architectures**
- Implemented various deep learning techniques, including CNNs, LSTMs, RNNs, VAEs, Transformers and attention to achieve multi-class classification accuracy of EEG signals for motor imagery tasks.
- Demonstrated the potential of deep learning techniques for EEG signal analysis by achieving a classification accuracy rate of 75% on the entire dataset.
- **Detecting Pulse from Head Movement:** I replicated the paper "Detecting Pulse from Head Movement" by Guha Balakrishnan, Fredo Durand, John Guttag.The code was able to detect the Heartbeat with an error of around 2-5%. The code was done in Python and it took around 2.5 weeks to complete

Publication

SIGGRAPH 2022 Blending camera and 77 GHz radar sensing for equitable, robust plethysmography

Extracurricular Activities

- **Eyantra 2016 Robotics Competition**: Coached a team of 3 students to the semi-finals in a National Level Robotics Competition. Managed and delegated tasks effectively to meet deadlines. Designed a robot with navigation capability and an arm for object pickup.
- Professional Bharatanatyam dancer (Top 1% in Senior Exam)
- Presented at IEEE Symposium Series on Computational Intelligence
- Member, Center for Intelligent Systems at PES University
- Operations Lead, Epsilon 2016 (2000+ participants, 100+ events)